



# GS2: A Gram-Schmidt Approach to Gene Selection for High-Dimensional Genomic Data

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## Abstract

High-dimensional genomic datasets present major challenges for identifying biologically meaningful features that enhance predictive modeling while remaining interpretable. This paper proposes an interpretable framework, Gram-Schmidt Gene Selection (GS2), to identify informative and minimally redundant genes through orthogonalization. The framework is evaluated on multiple benchmark gene expression datasets and compared with established feature selection methods. Results show that GS2 consistently improves classification performance while reducing feature redundancy, supporting clearer model interpretation. The biological relevance of selected genes is assessed through Gene Ontology (GO) enrichment analysis, revealing significant associations with pathways and functions consistent with known cellular mechanisms. Overall, the framework demonstrates that GS2 can effectively integrate predictive power with biological interpretability, offering a practical tool for selecting GO-informed genes in genomic research.

**Key words:** Principal Component Analysis, Gram-Schmidt Orthogonalization, Gram-Schmidt Gene Selection, Interpretable Dimensionality Reduction, Gene Ontology

## Introduction

High-dimensional genomic data often comprise tens of thousands of molecular features, including gene expression levels, single nucleotide polymorphisms (SNPs), and epigenetic markers, motivating the use of powerful dimensionality reduction techniques. Principal Component Analysis (PCA) is one of the most established and widely used approaches for this purpose. Since its development in the early 20th century, PCA has become a fundamental tool in statistical learning, pattern recognition, and data analysis [1, 7]. By identifying orthogonal directions of maximal variance and projecting the data onto a lower-dimensional linear subspace spanned by these principal components, PCA facilitates efficient representation of complex, high-dimensional data, improving computational tractability and enhancing interpretability in downstream analytical tasks. Consequently, PCA has become an indispensable exploratory tool in genomics [17]. A well-established application arises in genome-wide association studies (GWAS), in which PCA is routinely employed to detect and correct for population stratification, thereby mitigating confounding effects attributable to ancestry-related structure [16, 13]. In transcriptomic research, PCA is widely used to extract global expression patterns, reduce technical

variation, and reveal biologically meaningful sample groupings, such as tumor subtypes and differentiation states [3]. The rapid development of single-cell transcriptomic technologies has further underscored the importance of PCA, as it provides a key initial step for capturing dominant axes of cellular heterogeneity and facilitates the visualization of distinct cell populations and developmental trajectories. In epigenomic and proteomic studies, PCA similarly enables the identification of major sources of variation that correspond to interpretable biological processes, including cell cycle progression, tissue specificity, and responses to environmental perturbations [10].

PCA facilitates downstream analyses by reducing the dimensionality of omics datasets, thereby alleviating the curse of dimensionality and improving the performance and stability of statistical models [6]. Throughout this paper, we use the term “features” to refer specifically to “genes”. Principal components are frequently incorporated as features in regression frameworks to account for latent sources of heterogeneity, or used as input for clustering algorithms to enhance robustness and computational tractability. However, PCA has several important limitations. First, it is inherently a linear method and therefore cannot capture the nonlinear relationships that characterize gene regulatory networks, cellular signaling pathways, and other

complex biological processes [2, 19, 20, 18]. Second, PCA can perform poorly in settings where the number of genes greatly exceeds the number of samples (e.g. cells), because the sample covariance matrix becomes unstable and leading components may be dominated by noise rather than true biological signal [5, 14, 7]. Third, PCA struggles with highly sparse datasets such as single-cell transcriptomic data, where dropout events and zero inflation distort variance estimates and cause linear projections to misrepresent relationships among samples [15, 8, 11]. As a result, reliance solely on PCA may yield oversimplified or distorted representations of the underlying biological structure, motivating the use of more flexible dimensionality reduction methods that account for nonlinear structure, sparsity, and the statistical behavior of high-dimensional data [4, 6].

To address these limitations, we propose the Gram-Schmidt Gene Selection (GS2) algorithm. GS2 is an unsupervised algorithm that iteratively identifies and removes redundant genes using a Gram-Schmidt orthogonalization over chosen multilinear polynomials. At each step, GS2 computes a variance vector for the current transformed data, selects the gene with the largest remaining variance, and then orthogonalizes the function family with respect to this newly selected gene so that nonlinear dependencies can be removed before the next step. The process continues until the maximum remaining variance drops below a user-defined threshold, yielding a subset of genes that retains the informative structure of the data while discarding those that are linearly or nonlinearly redundant. GS2 is used as an add-on unit before applying PCA for downstream analyses. It is a feature selection method capable of capturing nonlinear dependencies between genes, because it applies a Gram-Schmidt orthogonalization over multilinear polynomials. This is particularly relevant in many genomics applications, where genes are often binary (e.g., mutation presence/absence, motif matches), and where multilinear polynomials naturally and expressively represent complex nonlinear interactions among such genes. It operates effectively even on sparse datasets, since GS2 relies only on expectations of  $\mathcal{L}^2$  functions, quantities that remain well-defined and stable for low-degree multilinear polynomials (e.g., degree 2) even under heavy sparsity. Finally, it is well-suited for high-dimensional settings, as its iterative evaluation of individual genes allows it to prune redundant dimensions early, reducing computational load, and its structure scales primarily with the size of the selected function family rather than the full data dimension, making it efficient and practical for large-scale genomic and other high-dimensional analyses. Note that GS2 differs fundamentally from variance-stabilizing (vst-based) approaches. Whereas vst focuses on modeling the mean-variance relationship for each gene independently, GS2 is explicitly designed to capture higher-order dependencies among genes. Our goal is to demonstrate that incorporating such higher-order structure leads to measurable improvements in downstream analyses.

Finally, to validate our theoretical results, we conduct extensive numerical experiments on benchmark datasets. To demonstrate that GS2 selects biologically meaningful genes, we apply GS2 to multiple benchmark datasets. We then perform PCA on the selected genes and compare the resulting gene ontology analysis with that obtained by applying PCA to the full dataset. Our results indicate that several genes highlighted when PCA is applied to the full dataset are irrelevant to the underlying biological processes. To further evaluate whether this improved informativeness leads to better classification performance, we

apply k-nearest neighbors (k-NN) in both scenarios and show that the results obtained after applying GS2 outperform those obtained without GS2. Finally, we compare GS2 with several well-known feature selection methods in the literature and show that its classification accuracy is significantly higher than that of existing algorithms.

## Method

In this section, we first introduce the Gram-Schmidt Gene Selection (GS2) method. We then describe the dataset used to evaluate the proposed algorithm, followed by a presentation of the experiments conducted. We present the results for a single benchmark dataset here, while the corresponding results for the remaining benchmark datasets are reported in the supplementary material. Finally, we outline our evaluation methodology, which assesses performance in terms of classification accuracy, gene ontology analysis, and the relevance of the selected genes to the benchmark dataset.

### Proposed Algorithm: GS2

#### Notation

Let  $X = [X_1, \dots, X_d]^\top \in \mathbb{R}^d$  denote a gene vector whose entries are real-valued (e.g., gene expression measurements) and sampled i.i.d. from an unknown distribution  $P_X$ . Deterministic vectors are written as  $\mathbf{x} = [x_1, \dots, x_d]^\top$ . We use  $[d] = \{1, \dots, d\}$  and, for  $j \leq d$ , set  $s_{[j]} = \{s_1, \dots, s_j\}$ . For any subset  $\mathcal{S} \subseteq [d]$ , the notation  $X_{\mathcal{S}}$  (resp.  $\mathbf{x}_{\mathcal{S}}$ ) refers to a vector of length  $|\mathcal{S}|$  which contains the entries of  $X$  (resp.  $\mathbf{x}$ ) indexed by  $\mathcal{S}$ ; likewise,  $X_{s_{[j]}}$  collects the entries of  $X$  in  $s_{[j]}$ . Let  $\mathcal{P}(\mathbf{x})$  be a family of multilinear polynomials in  $\{x_1, \dots, x_d\}$ . For  $\mathcal{S} \subseteq [d]$ , the set  $\mathcal{P}(\mathbf{x}_{\mathcal{S}})$  consists of those polynomials depending only on variables with indices in  $\mathcal{S}$ . When replacing  $x_i$  by their random counterparts, we write  $\mathcal{P}(X_{\mathcal{S}})$  for the result of substituting  $x_i \leftarrow X_{s_i}$  for  $\mathcal{S} = \{s_1 < \dots < s_{|\mathcal{S}|}\}$ . We use the algorithmic notation  $a \leftarrow b$  to denote that the expression  $b$  is computed, and then assigned into the variable  $a$ .

**Definition 1** A multilinear polynomial is a polynomial in which each variable appears with degree at most 1; equivalently, it is linear in every variable when all other variables are held constant. For example  $f(a, b, c) = 3a + 2bc - 5ac$  is multilinear because no variable appears with power greater than 1 in any term.

**Example 1** Let  $d = 4$  and  $\mathcal{P}(\mathbf{x}) = \{x_1, x_2, x_4, x_1x_4, x_2x_3x_4\}$ . For  $\mathcal{S} = \{1, 4\}$  we have  $\mathcal{P}(\mathbf{x}_{\mathcal{S}}) = \{x_1, x_4, x_1x_4\}$ , and for  $\mathcal{T} = \{2, 3\}$  we obtain  $\mathcal{P}(\mathbf{x}_{\mathcal{T}}) = \{x_2, x_3\}$ .

The random vector  $X$  used in this paper does not need to be specified explicitly. We treat random variables symbolically and use them only through their expectations. In practice, these expectations are approximated using empirical averages computed from the data. For example, one may define  $g(X) = 2X_1 - 3X_3$  without needing to know the distribution of  $X$ . To keep the notation simple, we write expressions such as  $\mathbb{E}[g(X)]$ , while implicitly understanding that, in applications, this quantity is estimated by the empirical mean  $\mathbb{E}[g(X)] \approx \frac{1}{N} \sum_{i=1}^N g(\alpha_i) = \frac{1}{N} \sum_{i=1}^N (2\alpha_{i,1} - 3\alpha_{i,3})$ , where  $\{\alpha_i\}_{i=1}^N$  denote the samples, with  $\alpha_i = (\alpha_{i,1}, \dots, \alpha_{i,d})^\top$  representing the  $i^{th}$  sample and  $\alpha_{i,j}$  its  $j^{th}$  entry.

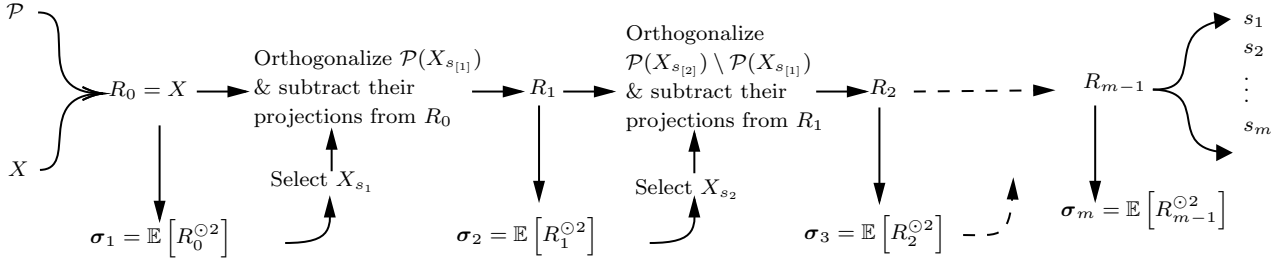


Fig. 1: **A schematic illustration of GS2.** The variance vectors  $\{\sigma_i\}_{i=1}^m$  are computed iteratively by subtracting projections onto the already orthogonalized functions in  $\mathcal{P}(\mathbf{x})$ , whose variables  $X_{s_i}$  have been fixed as the selected genes of  $X$ . Each iteration yields a new random variable  $R_j$  together with its corresponding variance vector  $\sigma_j$ . These variance vectors subsequently determine, at every step, the entry of  $R_j$  with the largest variance, which is then selected for the next iteration.

Our methods do not require any specific assumptions on  $\mathcal{P}(\mathbf{x})$  or on  $P_X$ , except that  $\mathcal{P}(X_S) \subseteq \mathcal{L}^2(P_X)$  for every  $S \subseteq [d]$ . In other words, after substituting  $x_i$  with  $X_{s_i}$ , each resulting function must have finite variance and therefore belong to the Hilbert space equipped with the inner product  $\langle f, g \rangle = \mathbb{E}[fg]$ . Throughout the paper, every random function is assumed to be a measurable function of the base random vector  $X$ , from which the data samples are drawn i.i.d., and all expectations are taken with respect to  $P_X$ .

### Gram-Schmidt Orthogonalization over Function Spaces

In this section, we explain the Gram-Schmidt (GS) process on function spaces, as summarized in Algorithm 1. We denote by  $\hat{\mathcal{P}}(X_S)$  the result of applying the Gram-Schmidt process over multilinear polynomials  $\mathcal{P}(X_S)$ . Throughout this paper, we make use of the well known Hadamard product, denoted by  $\odot$ , where for two vectors  $\mathbf{x} = (x_i)_{i=1}^n, \mathbf{y} = (y_i)_{i=1}^n$  we have  $\mathbf{x} \odot \mathbf{y} = (x_i y_i)_{i=1}^n$ ; similarly  $\mathbf{x}^{\odot 2} = (x_i^2)_{i=1}^n$ .

We fix multilinear polynomials  $\mathcal{P}(\mathbf{x})$  in variables  $\mathbf{x} = (x_1, \dots, x_d)$ . Our algorithm proceeds in an iterative manner: at iteration  $j$ , we select a new feature  $X_{s_j}$ , apply the substitution  $x_j \leftarrow X_{s_j}$  to the functions in  $\mathcal{P}(\mathbf{x}_{[j]})$ , and then perform GS orthogonalization. At each iteration  $j$ , Algorithm 1 starts with the previously orthogonalized multilinear polynomials  $\hat{\mathcal{P}}(X_{s_1}, \dots, X_{s_{j-1}})$ , together with the newly introduced functions in  $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$  and the new random variable  $X_{s_j}$ . For every function in  $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$ , the algorithm substitutes  $x_i \leftarrow X_{s_i}$  for all  $i \in [j]$  (Line 5), subtracts its projection onto the functions already orthogonalized (Line 6), normalizes the resulting function (Line 7), and appends it to  $\mathcal{T}$  (Line 8). After this projection step, the algorithm defines a new random vector  $R_j$  derived from  $X$  (Line 11), computes its variance vector and returns it as output (Line 12).

### Gram-Schmidt Gene Selection (GS2)

Algorithm 2 presents the *Gram-Schmidt Gene Selection* (GS2) algorithm. At iteration  $j$ , the algorithm examines the variance vector  $\sigma_j$  to determine which feature of  $R_{j-1}$  exhibits the largest variance. If this maximal variance is below the threshold  $\epsilon^2$ , the procedure halts. Otherwise, the feature of  $R_{j-1}$  with highest variance is selected and denoted by  $X_{s_j}$ . The algorithm then performs a Gram-Schmidt orthogonalization step on the multilinear functions that involve  $X_{s_{[j]}}$  but exclude any dependence only on  $X_{s_{[j-1]}}$ .

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#### Algorithm 1 GS( $\hat{\mathcal{P}}(X_{s_{[j-1]}}), \mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$ , $X_{s_{[j]}}$ )

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**Require:**

- $\hat{\mathcal{P}}(X_{s_{[j-1]}})$ : orthogonalized multilinear polynomials.
- $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$ : all the multilinear polynomials in  $\mathcal{P}(\mathbf{x})$  which depend on  $\mathbf{x}_{[j]}$  but not only on  $\mathbf{x}_{[j-1]}$ .
- $X_{s_{[j]}}$ : the random variables which appear in  $\hat{\mathcal{P}}(X_{s_{[j-1]}})$ , and a new random variable  $X_{s_{[j]}}$ .

**Ensure:** Orthogonalized multilinear polynomials  $\hat{\mathcal{P}}(X_{s_{[j]}})$  and a variance vector  $\sigma_{j+1}$ .

- 1: **Initialize:**  $\mathcal{T} \leftarrow \hat{\mathcal{P}}(X_{s_{[j-1]}})$ .
  - 2: Denote  $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]}) \triangleq \{f_1(\mathbf{x}), \dots, f_\ell(\mathbf{x})\}$ ,
  - 3: where  $f_1(\mathbf{x}) = x_j$ .
  - 4: **for**  $k \leftarrow 1$  **to**  $\ell$  **do**
  - 5:    $g(X_{s_{[j]}}) \leftarrow f_k(X_{s_{[j]}})$
  - 6:    $g(X_{s_{[j]}}) \leftarrow g(X_{s_{[j]}}) - \sum_{f \in \mathcal{T}} \mathbb{E}[gf]f$
  - 7:    $g(X_{s_{[j]}}) \leftarrow \frac{g(X_{s_{[j]}})}{\sqrt{\mathbb{E}[g(X_{s_{[j]}})^2]}}$
  - 8:    $\mathcal{T} \leftarrow \mathcal{T} \cup \{g(X_{s_{[j]}})\}$
  - 9: **end for**
  - 10: Define  $\hat{\mathcal{P}}(X_{s_{[j]}}) = \mathcal{T}$ .
  - 11: Define  $R_j = X - \sum_{f \in \mathcal{T}} \mathbb{E}[Xf]f$ .
  - 12: Define  $\sigma_{j+1} = \mathbb{E}[R_j^{\odot 2}]$ .
  - 13: Return  $\sigma_{j+1}, \mathcal{T}$ .
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#### Algorithm 2 Gram-Schmidt Gene Selection (GS2)

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**Require:** Multilinear polynomials  $\mathcal{P}(\mathbf{x})$  and a threshold  $\epsilon > 0$ .

**Ensure:** Selected features  $s_1, \dots, s_m$  ( $m$  is a varying number that depends on  $\mathcal{P}(\mathbf{x})$ ,  $\epsilon$ , and  $X$ )

- 1: **Initialize:**  $\sigma_1 = \mathbb{E}[X^{\odot 2}]$ .
  - 2: **for**  $j \leftarrow 1$  **to**  $d$  **do**
  - 3:   Let  $s_j \triangleq \arg \max_{i \in [d]} \{\sigma_{j,i}\}_{i=1}^d$ , where  $\sigma_j = (\sigma_{j,i})_{i=1}^d$ .
  - 4:   **if**  $\|\sigma_j\|_\infty \leq \epsilon^2$  **then**
  - 5:     break.
  - 6:   **end if**
  - 7:    $\sigma_{j+1}, \hat{\mathcal{P}}(X_{s_{[j]}}) = \text{GS}(\hat{\mathcal{P}}(X_{s_{[j-1]}}), \mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]}), X_{s_{[j]}})$
  - 8: **end for**
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### Dataset

In this study, we applied our methodology to five benchmark datasets. Specifically, in this section, we present the results for one representative dataset, while the corresponding analyses

for the remaining datasets are provided in the supplementary materials. We incorporated a combination of single-cell RNA sequencing and MERFISH (multiplexed error-robust fluorescence in situ hybridization) of the preoptic region in the mouse hypothalamus [12], which provides a comprehensive molecular, spatial, and functional survey of the hypothalamic preoptic region in *Mus musculus*. This dataset was generated by Moffitt *et al.* using high-throughput scRNA-seq on the Illumina NextSeq 500 platform, profiling approximately 31,000 dissociated cells collected from the preoptic area of multiple adult male and female mice. The experimental design includes six samples, comprising three biological replicates from pooled female mice and three biological replicates from pooled male mice, enabling robust comparisons of sex-specific transcriptional signatures. The work was produced by the Moffitt Laboratory at Boston Children’s Hospital and is publicly accessible through the Gene Expression Omnibus (GEO) under accession number GSE113576.

### Experimental Setup

To evaluate the effectiveness of our proposed GS2 method, we first applied GS2 to the dataset without using any class labels, consistent with the unsupervised nature of the algorithm. To implement GS2, we used multilinear polynomials of degree 2 in  $\mathcal{P}$  functions, and all components of the method were implemented in Python. After obtaining the subset of features selected by GS2, we analyzed the results from multiple perspectives, including gene ontology enrichment and visualization, classification accuracy, and comparisons with several well-known feature selection algorithms in the literature.

### Gene Ontology

We conducted a comparative analysis against the full, raw dataset to assess whether the selected features by GS2 capture biologically meaningful variation related to the underlying cell types. Specifically, we performed principal component analysis (PCA) on both the raw gene expression matrix and the reduced feature set produced by GS2, extracting the top 20 principal components in each condition. For each principal component, we identified 100 genes with the largest absolute coefficients, representing the features most influential in defining the major genes in each principal component. Using the PCA-derived gene sets, we categorized the features into three groups:

1. Genes shared between the PCA results of the dataset and those of the GS2-selected subset
2. Genes uniquely identified through PCA on the dataset
3. Genes uniquely identified through PCA on the GS2-selected subset.

We then perform gene ontology (GO) enrichment analysis using the R package Enrichr [9] on each group to determine their functional relevance and to assess whether GS2 preferentially retains genes associated with known cell-type specific biological processes. Gene set library “GO\_Biological\_Process\_2018” is used with default parameters. The combined score is defined as the P-value computed using the Fisher’s exact test multiplied with the z-score of the deviation from the expected rank. We also use Uniform Manifold Approximation and Projection (UMAP), implemented through the `umap-learn` package, to visualize the results. we employ PCA implementation from `sklearn.decomposition` module.

To show the importance of the results of GO, we divide them into three categories as follows:

- **Relevant:** Biological processes, pathways, or molecular functions that are directly intrinsic to the system or tissue under study, as well as those that, while not unique to it, are commonly active, functionally important, or contextually significant within it. Together, these terms encompass both the core cellular mechanisms, development, and physiology of the primary cell types of interest, and the supporting or regulatory biology that directly influences their function.
- **Weakly relevant:** Processes that are general, systemic, or peripheral to the main biology of the system, but that may appear due to cross-talk, infiltration, or shared molecular machinery. They are biologically valid but not central to the dataset’s focus.
- **Irrelevant:** Processes with no clear or plausible connection to the biology of the system being studied. These terms likely arise due to background noise, annotation artifacts, or statistical enrichment unrelated to the experimental context.

### Classification Accuracy

To compare the results quantitatively, we evaluated both the raw data and the features selected by GS2 by running a classification task after applying PCA. Finally, to assess how well the selected features support downstream predictive tasks, we measured classification performance using a k-Nearest Neighbors (k-NN) classifier trained on the resulting low-dimensional embeddings. This evaluation allowed us to quantify whether the unsupervised feature selection carried out by GS2 preserves the discriminative structure necessary for distinguishing cell types. we employed k-NN implementation from the `sklearn.neighbors` module in Python.

## Results

In this section, we present the numerical simulation results of GS2, which is described in detail in Section 2. Figure 2 illustrates a detailed schematic of the first set of experiments, including the overall workflow and evaluation procedures. These experiments are designed to assess the ability of GS2 to identify the most informative genes from high-dimensional data. Specifically, we report three complementary types of results: (1) classification accuracy, which evaluates the predictive performance of the selected genes; (2) gene ontology analysis, which examines the biological relevance and functional enrichment of the selected genes; and (3) visualization, which provides an intuitive interpretation of the gene selection behavior of GS2.

### Classification Accuracy

In this section, we compare the classification accuracy (%) of PCA applied to both the raw gene expression matrix and the reduced gene set generated by GS2. The quantitative results are summarized in Table 1. We consider three categories of data: (1) the entire dataset, comprising 31,299 cells; (2) excitatory neurons, comprising 3,511 cells; and (3) inhibitory neurons, comprising 15,042 cells. Across all three cases, the results reported in Table 1 demonstrate that applying GS2 consistently improves classification accuracy. Notably, even in the setting with a relatively small number of samples—namely, the excitatory neuron subset—GS2 yields strong classification performance,

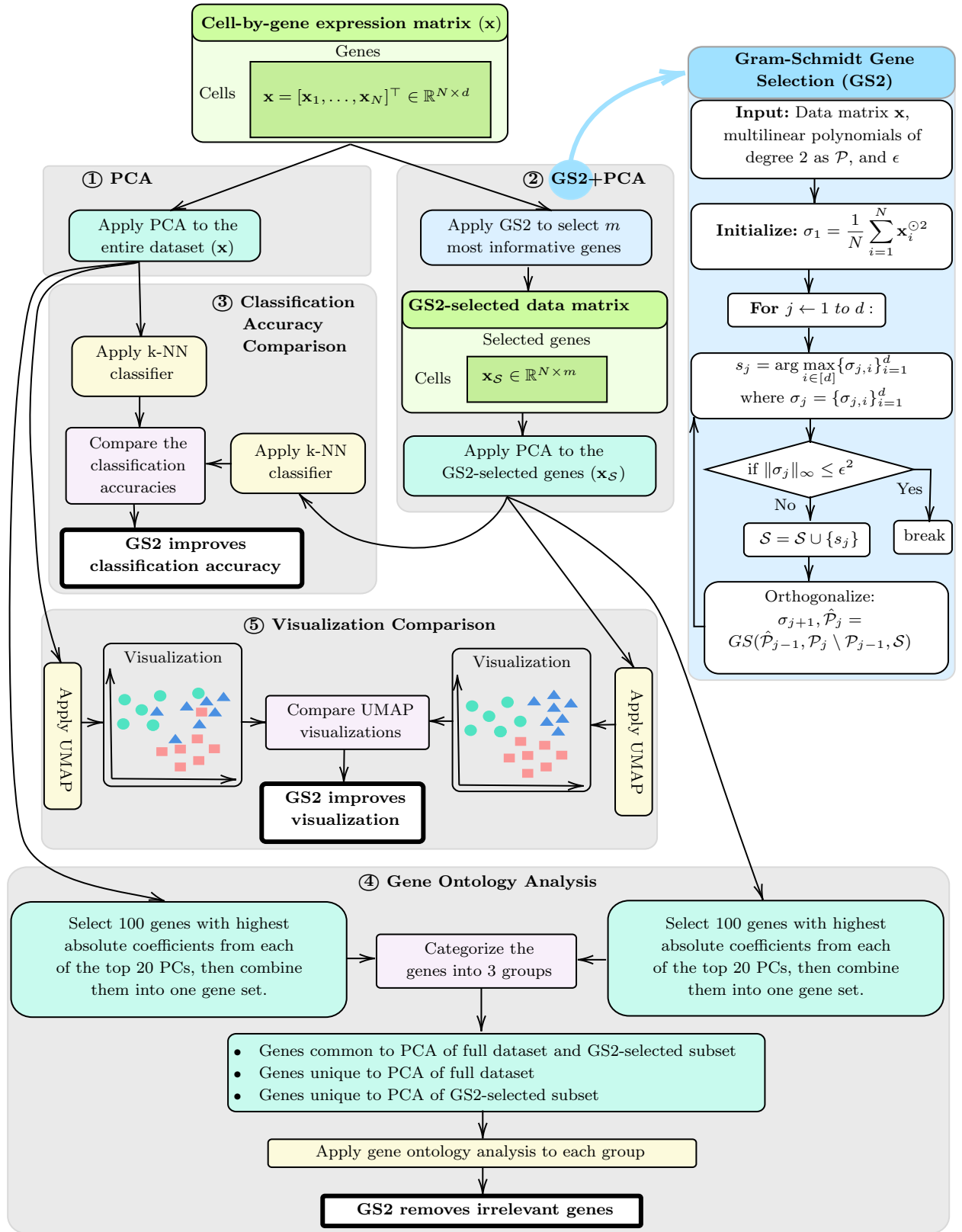


Fig. 2: A schematic illustration of the experimental setup.

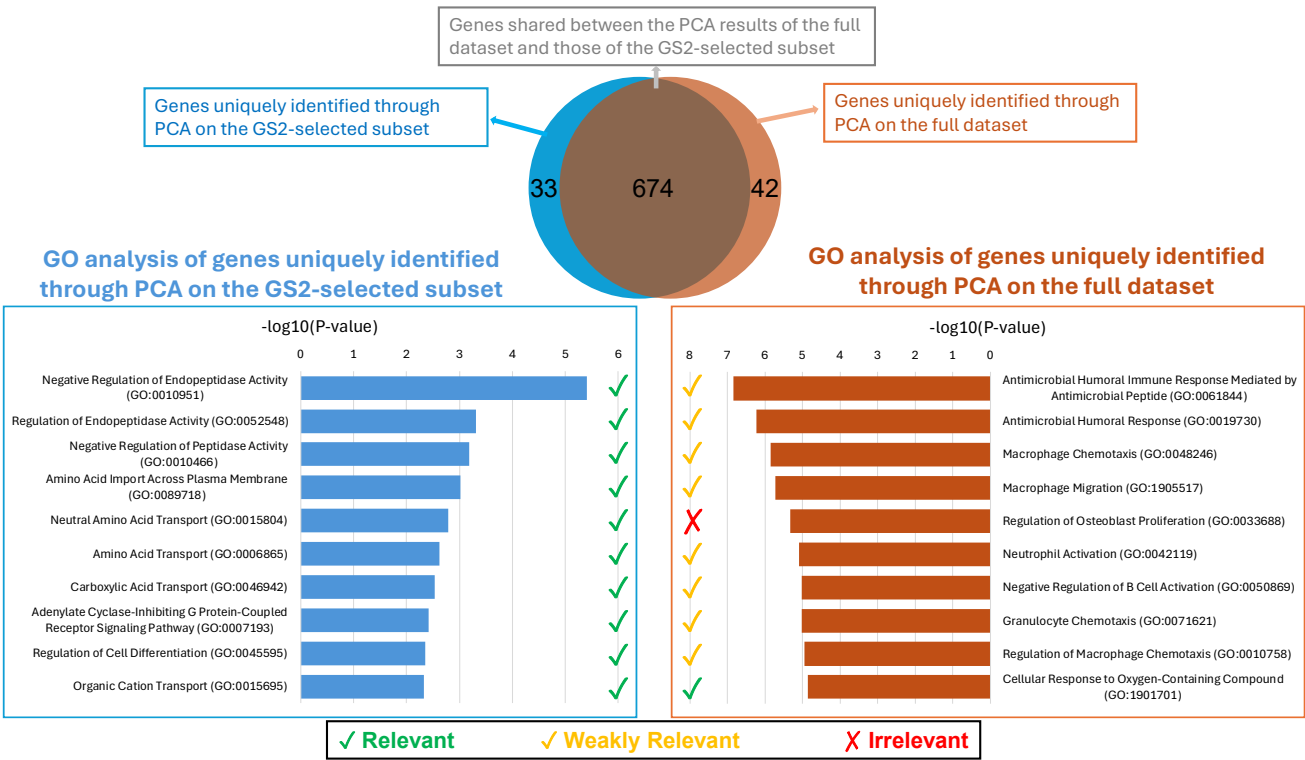


Fig. 3: Gene ontology enrichment analysis comparing PCA applied to GS2-selected genes and PCA applied to the full gene expression dataset. PCA on GS2-selected genes highlights pathways related to neural signaling, cellular differentiation, and metabolic regulation, whereas PCA on the full dataset yields predominantly immune-related and biologically irrelevant categories, illustrating the improved functional relevance achieved through GS2-based gene selection.

highlighting its effectiveness in low-sample regimes. Since the number of samples available for Excitatory and Inhibitory neurons is relatively small, the k-Nearest Neighbors (k-NN) classifier has limited representative data to learn from. As a result, the distances between samples become less reliable for distinguishing the two neuron types, which leads to reduced classification accuracy in these cases.

**Table 1.** Comparison of classification accuracy (%) of PCA using the raw gene expression matrix and the reduced gene set obtained via GS2 for the full dataset, excitatory neurons, and inhibitory neurons.

| Dataset  | Entire dataset | Excitatory | Inhibitory |
|----------|----------------|------------|------------|
| GS2      | 90.70          | 57.33      | 51.97      |
| Raw Data | 86.67          | 50.36      | 48.45      |

Gene Ontology Analysis

In this subsection, we investigate the biological interpretability of the selected genes through gene ontology analysis detailed in section 2.3.1. By comparing PCA applied to GS2-selected genes with PCA applied to the entire gene expression dataset, we assess how GS2 influences the functional relevance of the resulting gene sets. This analysis provides insight into the ability of GS2 to prioritize biologically meaningful signals while suppressing

spurious or non-informative patterns. Figure 3 highlights distinct differences between the biological processes identified by GS2.

When PCA was applied to the GS2-selected genes, the resulting gene set emphasized pathways directly related to neural and metabolic regulation. Several enriched categories were associated with amino acid and carboxylic acid transport (GO:0089718, GO:0015804, GO:0006865, GO:0046942), processes that are essential for neurotransmitter precursor uptake, glutamate/GABA cycling, and metabolic coupling between neurons and glia. Additional terms highlighted inhibitory G protein-coupled receptor (GPCR) signaling (GO:0007193) and the regulation of cell differentiation (GO:0045595), both of which play central roles in hypothalamic signal transduction and lineage specification. Enrichment of processes involved in the regulation of endopeptidase and peptidase activity (GO:0010951, GO:0052548, GO:0010466) further underscored mechanisms controlling peptide hormone activity and extracellular signaling. Collectively, these results indicate that PCA on GS2-selected genes captures pathways that reflect core aspects of hypothalamic function, including neurotransmission, cellular differentiation, and metabolic regulation.

In contrast, when PCA was applied to the entire gene expression dataset, the resulting gene set exhibited a strong bias toward immune-related categories. Enriched terms included antimicrobial humoral immune responses (GO:0061844, GO:0019730), macrophage chemotaxis and migration (GO:0048246, GO:1905517), and neutrophil or granulocyte activation and

chemotaxis (GO:0042119, GO:0071621). Although these pathways may reflect microglial activity or immune infiltration signatures, they are not central to hypothalamic biology. Moreover, several enriched categories, such as regulation of osteoblast proliferation (GO:0033688), were biologically irrelevant to neural tissue, suggesting spurious enrichment. Only a limited subset of terms, including cellular responses to oxygen-containing compounds (GO:1901701), showed partial relevance to neural or metabolic processes.

Taken together, these findings demonstrate that GS2 identifies gene sets that map to biologically coherent pathways central to hypothalamic and neuronal function. This reinforces the advantage of GS2 in prioritizing functionally meaningful genes in complex single-cell datasets.

### Visualization

In this subsection, we focus on visualization using Uniform Manifold Approximation and Projection (UMAP) to assess the effect of GS2-selected genes. To provide a more challenging and informative evaluation, we restrict the analysis to the ten cell types with the smallest number of samples, as these rare cell populations are typically difficult to distinguish due to limited representation and high overlap in expression space. When using the full dataset, the resulting UMAP embeddings exhibit substantial crowding, making it difficult to visually assess differences in cluster separation. By concentrating on these underrepresented cell types, we create a more stringent visualization setting in which improvements in clustering structure are more readily observable. As illustrated in Figure 4, GS2-selected genes yield clearer separation among these rare cell types compared to embeddings constructed from the full gene set, highlighting the advantage of GS2 in enhancing discriminative structure under challenging conditions.

The circled regions in Figure 4 highlight representative cell types whose separation is markedly improved when GS2-selected genes are used. In the GS2-based embedding, these cell populations form compact and well-defined clusters with clear boundaries, indicating that GS2 preserves discriminative gene expression patterns relevant to cell-type identity. In contrast, the corresponding regions in the embedding constructed from the entire gene set show substantial dispersion, elongation, or partial overlap with neighboring clusters, obscuring class boundaries. This effect is particularly pronounced for rare cell types, where noise introduced by non-informative genes can dominate the low-dimensional structure. Overall, these observations demonstrate that GS2 enhances the clarity of UMAP representations by amplifying meaningful variation and reducing feature-induced distortion, resulting in more visually separable and interpretable clusters.

### Conclusion

This paper introduces the Gram-Schmidt Gene Selection (GS2) method as a powerful tool for identifying informative genes from high-dimensional genomic data. Our extensive evaluation demonstrates that GS2 consistently outperforms traditional feature selection methods by improving classification accuracy and reducing redundancy. The method's effectiveness is highlighted across various datasets and experimental settings, including low-sample regimes. Additionally, gene ontology analysis confirms

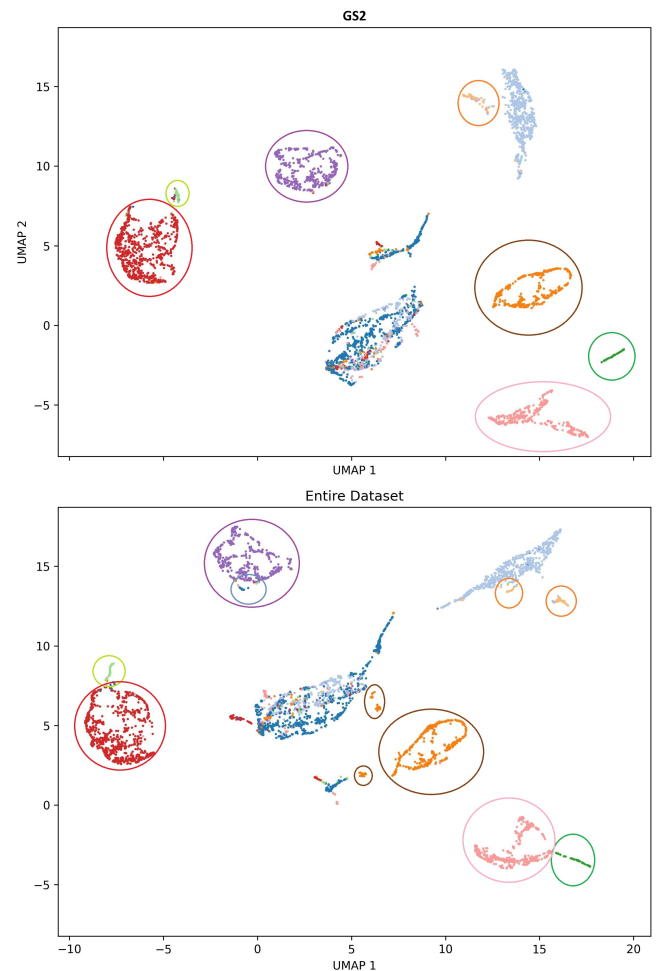


Fig. 4: UMAP visualization comparing GS2-selected genes with the full gene expression dataset.

the biological relevance and functional enrichment of the selected genes, further validating the utility of GS2 in genomic studies. These results underscore the potential of GS2 to enhance our understanding of gene regulation and cellular functions, making it a valuable resource for both research and clinical applications.

### Competing interests

The authors declare that they have no competing interests.

### Author contributions statement

B.Y. conceptualized the study, curated the data, performed the formal analysis, carried out the investigation, developed the methodology, administered the project, provided resources, developed the software, validated the findings, generated the visualizations, drafted the original manuscript, and wrote and reviewed the final version. B.S. developed the methodology, provided funding, administered the project, provided resources, supervised the work, and wrote and reviewed the final version. Y.E.L. conceptualized the study, curated the data, performed the formal analysis, carried out the investigation, developed the

methodology, provided resources, supervised the work, validated the findings, drafted the original manuscript, and wrote and reviewed the final version.

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